Sheet 37 Thu. 28 Jan. 2027 Quotient equation $\frac{A(x)}{B(x)} = k$

1. Warm-up · Flash [EQ] For which value of x the expression $\dfrac{1}{x-3}$ is it not defined?

Ans. $x = 3$.

2. Core · Method [EQ] Solve $\dfrac{2x-1}{x+3} = 0$.

Ans. $x \neq -3$ and $2x - 1 = 0 : x = \dfrac{1}{2}$.

3. Core + · Method [EQ] Solve $\dfrac{x+1}{x-2} = 3$.

Ans. $x \neq 2 ; x + 1 = 3(x-2)$, i.e. $x = \dfrac{7}{2}$. Check: $4.5\ \overline{1{,}5} = 3$.

4. Back · Expand [DEV] Expand and simplify $(x+2)(x-5)$.

Ans. $x^2 - 3x - 10$.

5. Running thread · Reading code [ALG] What do $g(3)$ and $g(-5)$?

```
def g(x):
    if x > 0:
        return 2*x
    else:
        return -x
```

Ans. $g(3) = 6$ and $g(-5) = 5$.

Condition evaluated before the calculation: the branch depends on the sign of x.

Sheet 38 Mon. 1st Feb. 2027 Sign of a product or quotient; sign tables

1. Warm-up · Flash [EQ] Sign of $2 \quad x - 4$?

Ans. Negative for $x < 2$, zero at 2, positive for $x > 2$.

2. Core · Method [EQ] Draw the sign table of $(x-1)(x+3)$.

Ans. Roots -3 and 1: positive on $]-\infty;-3[$, negative on $]-3;1[$, positive on $]1;+\infty[$.

Sign of the product: sign rule column by column.

3. Core + · Method [EQ] Solve $\dfrac{x+2}{x-1} \geqslant 0$.

Ans. Positive numerator for $x \geqslant -2$, positive denominator for $x > 1 ; S =]-\infty;-2] \cup]1;+\infty[$ (1 is excluded).

4. Back · Flash [PRB] 32-card deck: probability of drawing a heart? a king?

Ans. $\dfrac{8}{32} = \dfrac{1}{4} ; \dfrac{4}{32} = \dfrac{1}{8}$.

5. Running thread · Refute [LIT] Disprove: "for every real number x, $x^2 \geqslant x$ ».

Ans. Counterexample: $x = \dfrac{1}{2} : x^2 = \dfrac{1}{4} < \dfrac{1}{2} = x$.

A single counterexample is enough to disprove a universal statement.

Proportions and changes

Sheet 39 Tue. 2 Feb. 2027 Proportions and percentages; proportion of a proportion

1. Warm-up · Flash [PRO] Calculate 15% of 240.

Ans. $0{,}15 \times 240 = 36$.

2. Core · Method [PRO] 30% of pupils are boarders and 40% of boarders dance. Which proportion of the students are boarders who dance?

Ans. $0{,}3 \times 0{,}4 = 0{,}12 = 12\%$: a proportion of a proportion (multiply).

3. Core + · Problem [PRO] A lycée has 800 students, 55 % of whom are girls; 20 % of the girls take Latin. How many girls take Latin?

Ans. $800 \times 0{,}55 = 440$ girls; $440 \times 0{,}20 = 88$.

4. Back · Flash [QP] 3 notebooks cost 4.50 €. Price of 7 notebooks?

Ans. $1{,}50 \times 7 = 10{,}50$ €.

5. Running thread · True/false [EQ] True or false? Justify. The number -2 is a solution of the equation $3x + 1 = -5$.

Ans. True. $3 \times (-2) + 1 = -6 + 1 = -5$.

Sheet 40 Thu. 4 Feb. 2027 Absolute change, multiplier, rate of change

1. Warm-up · Flash [PRO] A price goes from 80 Ar to 92 Ar. What is the absolute change?

Ans. $92 - 80 = 12$ Ar.

2. Core · Method [PRO] For the same change, give the rate of change and the multiplier.

Ans. $t = \dfrac{12}{80} = 0{,}15 = 15\,\%$; $CM = 1 + t = 1{,}15$.

3. Core + · MCQ [PRO] A 20 % decrease corresponds to the multiplier:

A. 0,2 **B.** $-0{,}8$ **C.** 1,2 **D.** 0,8

Ans. D. 0,8

$CM = 1 + t = 1 - 0{,}20$.

4. Back · Flash [STA] Temperatures -2, 0, 3, 5, 9 (in °C). Mean and range?

Ans. Mean 3 °C; range $9 - (-2) = 11$ °C.

5. Running thread · Flash [SOL] Calculate the volume and total surface area of a cube with edge 3 cm.

Ans. $V = 3^3 = 27\ \text{cm}^3$; $A = 6 \times 3^2 = 54\ \text{cm}^2$.

Sheet 41 Mon. 8 Feb. 2027 Successive percentage changes

1. Warm-up · Flash [CAL] Calculate $1.2 \times 0{,}9$.

Ans. 1,08.

2. Core · Method [PRO] A 20% increase is followed by a 10% decrease. What is the overall rate?

Ans. $1{,}2 \times 0{,}9 = 1.08$: overall increase of 8%.

Rates do not add; it is the multipliers that multiply.

3. Core + · True/false [PRO] True or false? Justify. A 50 % rise followed by a 50 % fall brings back to the initial value.

Ans. False. $1.5 \times 0{,}5 = 0.75$: overall decrease of 25%.

4. Back · Flash [SUB] Calculate $x^2 - x$ for $x = \dfrac{1}{2}$.

Ans. $-\dfrac{1}{4}$.

5. Running thread · Write up [DEM] $ABCD$ is a parallelogram such that $AC = BD$. What type is it? Justify in one sentence.

Ans. Rectangle: a parallelogram whose diagonals have the same length is a rectangle.

Sheet 42 Tue. 9 Feb. 2027 Reverse percentage change; proportion and percentage change

1. Warm-up · Flash [CAL] Calculate $\dfrac{1}{1{,}25}$.

Ans. 0,8.

2. Core · Method [PRO] After an increase of 25%, which decrease returns to the initial value?

Ans. $CM_{\text{rec}} = \dfrac{1}{1{,}25} = 0.8$: decrease by 20% (not 25%).

Pitfall: believing that an increase of 25 % is offset by a decrease of 25 %.

3. Core + · Interpret [PRO] Of 250 candidates, 150 are admitted; last year there were 200 candidates. Calculate the proportion admitted, then the rate of change of the number of candidates.